

# Muhammad Umer Bilal

mumberbilal142@gmail.com • +971 559695281 • Dubai, UAE •  
linkedin.com/in/muhammad-umer-bilal-b1a9391a2

---

## Statement of Motivation

Dear Professor [PROFESSOR LAST NAME],

I am writing to express my interest in pursuing a PhD under your supervision at [UNIVERSITY NAME], having read your work on [SPECIFIC PAPER OR RESEARCH AREA — e.g. "adaptive intrusion detection in IoT networks"]. My background combines a research foundation in machine learning for network security with over six years of hands-on experience building production backend and cloud systems, and I believe this combination would let me contribute meaningfully to your group's work.

I completed my MS in Computer Networks & Security at the National University of Computer & Emerging Sciences (NUCES-FAST), Islamabad, where my thesis proposed a dynamic window-sizing approach for detecting low-rate DDoS attacks using an ensemble of machine learning classifiers (RNN, MLP, LSTM, Random Forest), evaluated on the UNSW-NB15 dataset. This work is attached, along with my CV, as a writing sample.

Alongside this research background, I have spent the past six years working as a software engineer in Dubai, building backend systems, APIs, and cloud infrastructure for production applications. More recently, I have been deepening my practical security skills through hands-on web application penetration testing (PortSwigger Web Security Academy, Burp Suite) and am currently preparing for the eJPT certification. I see this practical, systems-level experience as a genuine complement to academic research — it means I can implement and stress-test ideas, not just model them theoretically.

*[OPTIONAL 2-3 SENTENCES: state specifically what you would want to research in their group and how it extends or connects to their recent publications — this is the paragraph that should change most per professor.]*

I would welcome the opportunity to discuss how my background might fit into your current research, and whether a funded PhD position or research assistantship may be available in your group. I have attached my CV, thesis writing sample, degree certificate, and reference letters from my thesis supervisor and course instructors for your reference.

Thank you very much for your time and consideration.

Sincerely,  
Muhammad Umer Bilal